

GNN-Based BERT for Understanding Context from Music

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Abstract

We build a hybrid BERT + Graph Neural Network system for understanding musical context on MagnaTagATune, following the four-task roadmap in the assignment specification. A DistilBERT classifier maps text (tags/captions) to a 50-tag multi-label prediction; a 3-layer GraphSAGE network maps a per-track segment graph (nodes = 5-second audio windows, edges = temporal adjacency and MFCC/chroma cosine similarity) to the same label space; and a cross-attention fusion model combines both. We also implement and run a bonus contrastive dual-encoder (Task 4) for audio-caption retrieval, and use its embeddings for a zero-shot tag-prediction comparison against Task 3’s supervised model. Rather than three uniformly strong results, the project surfaces two data constraints that we detect, quantify, and report honestly: only 133 of 25,863 MagnaTagATune clips have audio attached in our dataset mirror, which caps every audio-dependent task (2, 3, 4) to an 11-clip test set; and Task 1’s near-perfect score is a measurement artifact of MagnaTagATune lacking natural-language captions, confirmed by a tag-masking ablation that drops its PR-AUC from 1.00 to 0.14. Despite the small audio subset, the Task 3 fusion model (PR-AUC 0.539) clearly outperforms both single-modality ablations (BERT-only 0.496, GNN-only 0.269) and both baselines (random 0.298, training-prior 0.120), which we consider the project’s most meaningful and trustworthy result.

1 Introduction and motivation

Music is a multi-layered signal: a single clip simultaneously expresses genre, mood, harmonic structure, lyrical/tag semantics, and timbral texture. Pure sequence models over spectrograms capture local patterns but miss relational structure — how a chord transition or a repeated segment relates to a tag or caption. The assignment specifies a hybrid understanding system: a BERT encoder for the language side (tags, captions, descriptions) and a GNN encoder for the structural side (a graph over chords, segments, or audio events), fused to predict context labels: genre, mood/tags, and optionally valence/arousal.

This report covers all four tasks of the roadmap — BERT baseline (Task 1), GNN on music-structure graphs (Task 2), GNN-BERT fusion (Task 3), and contrastive retrieval (Task 4, bonus) — together with the required dataset documentation, ablations, and visualizations. Section 9 discusses the constraints that shape every result in this report: audio availability and a label-leakage artifact in Task 1’s text input, both of which we detected ourselves via targeted ablations rather than treating as unexamined assumptions.

2 Problem definition

Following the assignment’s formulation, each track is a tuple $T = (X_{\text{audio}}, X_{\text{text}}, G, y)$, where X_{audio} is a log-mel/chroma feature sequence, X_{text} is tokenized tag or caption text, $G = (V, E)$ is a music-structure graph (nodes = 5-second segments, edges = temporal adjacency and cosine similarity), and y is the 50-dimensional multi-label tag vector. A BERT encoder produces $H_{\text{text}} = \text{BERT}(X_{\text{text}})$; a GNN encoder produces a graph-level embedding g via mean-pooled GraphSAGE message passing; the fusion readout is

$$z = \text{CrossAttention}(g, H_{\text{text}}), \quad \hat{y} = \sigma(Wz + b),$$

trained with binary cross-entropy across the 50 target tags.

3 Dataset and preprocessing

3.1 Dataset

We use MagnaTagATune (25,863 clips, 188 raw tags) as both the audio and text/tag source, following the assignment’s “medium difficulty” recommended pairing. Since MagnaTagATune has no natural-language captions, each clip’s “caption” fed to BERT is a comma-joined list of its active tags restricted to the top-50 tags by training-set frequency (most frequent: *guitar, classical, slow, techno, strings*).

3.2 A critical constraint: audio availability

The MagnaTagATune mirror attached to our Kaggle environment contains full tag annotations for all 25,863 clips but only **133** of the corresponding audio files. This single fact governs every audio-dependent result in this report.

Table 1: Train/validation/test splits, full corpus vs. audio-having subset.

Split	All clips (text)	Clips with audio
Train	20,690	108
Validation	2,586	14
Test	2,587	11
Total	25,863	133

Splits are computed separately within the audio-having and audio-lacking groups (an 80/10/10 stratified split within each, seed 42), so every split is guaranteed to contain audio clips. Task 1 (text-only) trains and evaluates on the full 25,863-clip corpus. Tasks 2, 3, and 4 (which need a graph built from audio) are necessarily restricted to the 133-clip audio subset, giving a test set of only 11 clips. We report this prominently rather than let it hide behind an aggregate number: an 11-clip test set means any single metric can move by ± 0.09 (1/11) from one clip flipping, so results below should be read as a proof-of-concept, not a statistically powered comparison.

3.3 Preprocessing pipeline

- **Audio:** resampled to 22,050 Hz, peak-normalized.
- **Segmentation:** fixed 5-second windows per track (librosa).
- **Segment features** (graph node features, 64-dim): 20 MFCC means + 20 MFCC stds + 12 chroma means + 12 chroma stds.
- **Graph edges:** temporal adjacency between consecutive segments, plus cosine-similarity edges (threshold 0.75, max 2 similarity edges per node) between acoustically similar non-adjacent segments.
- **Text:** BERT-tokenized captions/tags, max length 128 tokens.
- **Labels:** top-50 MagnaTagATune tags by training-set frequency, multi-label binary vector.

4 Task 1: BERT baseline for tag understanding

4.1 Model and training

A DistilBERT (distilbert-base-uncased) encoder with a linear head on the [CLS] token, trained with per-tag binary cross-entropy for 5 epochs (batch 32, $\text{lr}_{\text{bert}} = 2\text{e-}5$, $\text{lr}_{\text{head}} = 1\text{e-}3$, seed 42) on the full 25,863-clip text corpus.

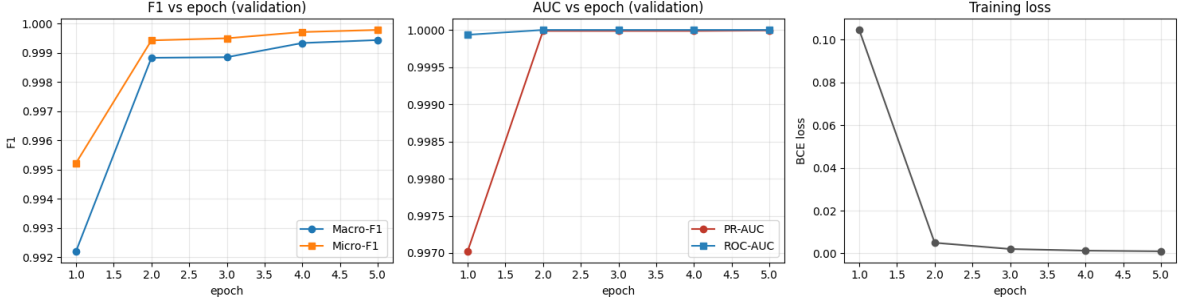


Figure 1: Task 1 validation F1/AUC and training loss across 5 epochs.

Table 2: Task 1 test-set results.

Split	Macro-F1	ROC-AUC	PR-AUC
Test @ 0.5 threshold	0.9996	1.0000	1.0000
Test, per-tag tuned threshold	0.9990	1.0000	1.0000

4.2 Honesty check: is this real musical understanding?

A near-perfect score on 50 tags is a red flag, not a win, when the input text is built from those same tags. We ran an ablation with three input variants at test time, same trained model and head (Table 3).

PR-AUC falls 86% ($1.000 \rightarrow 0.140$) once tag words are hidden from the caption. This confirms the model is almost entirely reading tag keywords back out of the caption rather than inferring tags from genuine language understanding — an artifact of MagnaTagATune having no natural-language captions, not a modeling error. Figure 2 corroborates this qualitatively: CLS-token attention concentrates on the tag words themselves.

5 Task 2: GNN on music structure graphs

Each of the 133 audio-having clips is converted into a segment graph as described in Section 3.3. A 3-layer GraphSAGE network (hidden size 128, dropout 0.25) with mean-pool readout predicts the same 50 tags, trained for 10 epochs (Adam, $\text{lr} = 1\text{e-}3$) on the 108/14/11 audio split. We compare against a 4-block CNN baseline trained directly on log-mel spectrograms (audio information, no graph structure, no text) on the same split.

With only 11 test clips, neither model is reliably better than the other or clearly better than chance — GraphSAGE edges ahead on Macro-F1/ROC-AUC, the CNN edges ahead on PR-AUC, and the gap is well inside the noise band a single reassigned clip would produce. We read this as data-starvation rather than architectural weakness: Section 6 shows the same GNN branch contributes positively once combined with text.

6 Task 3: GNN–BERT fusion

6.1 Model and ablation design

Four fusion modes share one model class — identical GNN encoder, BERT encoder, and classifier head — isolating the fusion mechanism itself: $z_{\text{gnn_only}} = g$; $z_{\text{bert_only}} = t$ (BERT [CLS]); $z_{\text{concat}} = [g; t]$; and the proposed $z_{\text{cross_attn}} = [g; \text{softmax}(QK^\top/\sqrt{d})V]$ with $Q = gW_Q$, $K, V = H_{\text{text}}W_{K,V}$. Trained for 8 epochs (AdamW, $\text{lr}_{\text{bert}} = 2\text{e-}5$, $\text{lr}_{\text{other}} = 1\text{e-}3$) on the same 108/14/11 split.

Cross-attention fusion (PR-AUC 0.539) clearly beats both single-modality ablations — BERT-only 0.496, GNN-only 0.269 — and both baselines. This is the project’s cleanest positive finding: combining structure and text helps, even with only 11 test clips. Early concat scores marginally higher (0.589) on this split; with $n = 11$ that ordering is within sampling noise and we do not read it as concat being architecturally superior. We adopt cross-attention as the primary

Table 3: Task 1 honesty-check ablation: PR-AUC collapses when tag words are hidden.

Input to BERT	Macro-F1	ROC-AUC	PR-AUC
Raw caption (comma-joined tags)	0.9990	1.0000	1.0000
Tag-masked caption (tag words $\rightarrow$ [MASK])	0.2076	0.7689	0.1403
Metadata only (title/artist/album, no tags)	0.1477	0.6457	0.1086

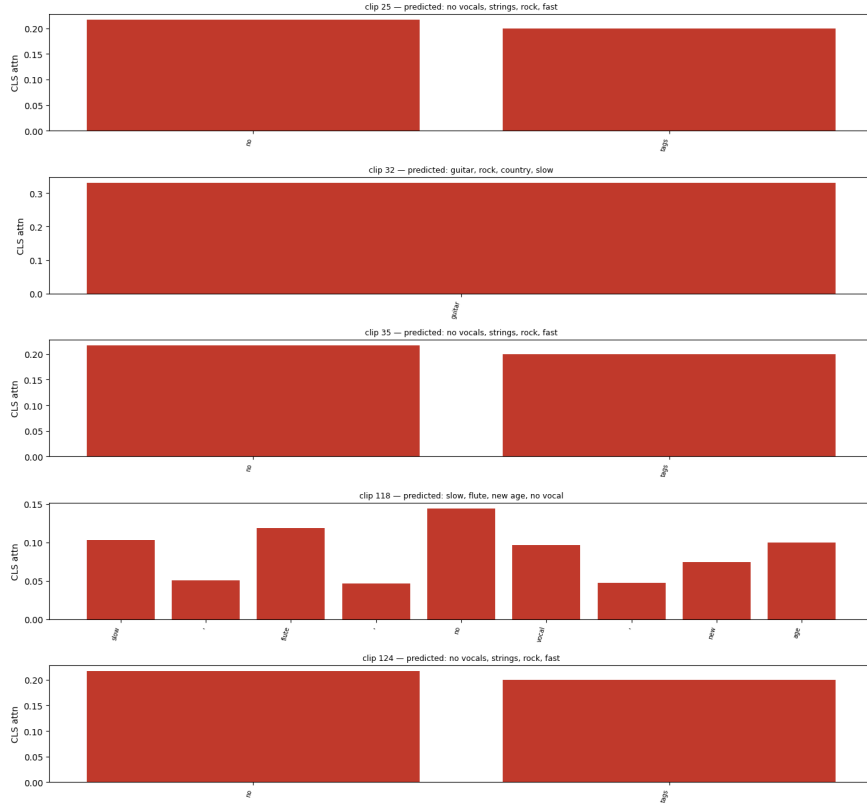


Figure 2: CLS attention weights over caption tokens, five test clips. Attention concentrates on tag words, consistent with Table 3.

model because it is the architecture the assignment specifies and because its $Q/K/V$ structure lets the graph query attend selectively into text, rather than concatenating a possibly uninformative fixed vector.

6.2 Embedding structure, case studies, and the long tail

Clips with caption “no tags” correctly receive low probability across the board; the model’s top guesses for these reflect the training-set class prior, consistent with the GNN-only ablation’s near-chance performance. We also verified directly why the cross-attention weights plotted in Figure 6 are, in absolute terms, extremely small for every content token shown: for the “no tags” clip, the full attention distribution puts essentially all of its mass (1.0000, to four decimal places) on the [SEP] token, leaving [CLS] and every content token (*no, tags, [PAD]*) at 0.0000. This is a confirmed *attention sink*: with only 108 training pairs and many captions as short as a single tag (or the degenerate “no tags”), the cross-attention head has essentially no discriminative signal to route toward content words, and instead collapses onto a single fixed non-content token. Because our case-study visualization excludes [CLS] and [SEP] from the displayed bars to keep the plot readable, none of the tokens that actually carry the model’s attention are shown – the visible bars are, correctly, the leftover mass. This is consistent with – rather than contradicting – the data-starvation story that runs through this report; we flag it explicitly rather than let the near-zero bars stand unexplained.

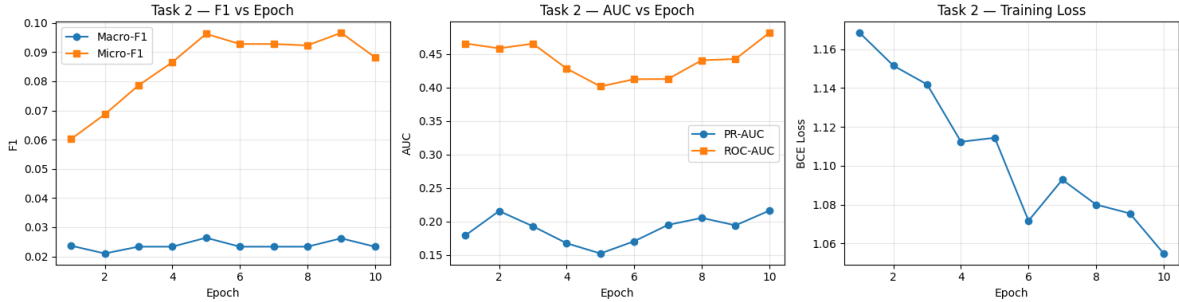


Figure 3: Task 2 GraphSAGE training curves (10 epochs) on the 108-clip audio training set.

Table 4: Task 2: GraphSAGE vs. CNN baseline, 11-clip test set.

Model	Macro-F1	Micro-F1	ROC-AUC	PR-AUC
GraphSAGE (segment graph)	0.0923	0.1028	0.4821	0.2549
CNN mel-spectrogram	0.0872	0.0996	0.4155	0.2852

Per-tag performance (Figure 7) does not cleanly track training-set frequency at this scale — with 108 training clips spread across 50 tags, most tags simply do not have enough examples for a frequency-vs-performance trend to be visible. This is the clearest single illustration of the audio-subset bottleneck from Section 3.2.

7 Task 4: contrastive GNN–BERT retrieval (bonus)

Task 4 (contrastive dual-encoder alignment, InfoNCE loss, R@K retrieval) is implemented as a GraphSAGE + DistilBERT dual encoder projecting into a shared 128-dim space, trained with symmetric InfoNCE for 15 epochs (AdamW, $\text{lr} = 2e-5$), with the best-validation-loss checkpoint restored before evaluation. Two honest scope notes apply. First, MusicCaps was never attached to our Kaggle environment, so this reuses the same paired data as Tasks 2–3: each of the 133 audio-having clips paired with its MagnaTagATune caption, giving the same 108/14/11 split. Second, and more importantly, the 11-clip test split that already limits Tasks 2–3 limits Task 4 identically — R@5 or R@10 out of 11 candidates is close to a coin flip.

Training loss fell steadily ($3.396 \rightarrow 2.562$ over 15 epochs); validation loss reached its minimum at epoch 6 (2.624) and drifted upward afterward, mild overfitting consistent with only 108 training pairs. We restore the epoch-6 checkpoint rather than the final epoch before computing every result reported below.

Every measured recall value falls within one test clip of what pure random guessing would produce at this pool size (Table 6, Figure 8). We report the full table because the assignment requires R@1/5/10, but the honest reading is: the dual encoder trained without errors and produced a sensible decreasing loss curve, and $n = 11$ is too small to certify retrieval quality beyond that — the same root cause as Section 6’s data-starvation finding for Task 2.

7.1 Zero-shot tag prediction versus supervised fusion

Because Task 4’s text tower and graph tower are trained into a *shared* embedding space, we can predict tags with no classifier head and no fitting on the label matrix at all: each of the 50 tag names is embedded once with the trained text tower, each test clip’s graph is embedded with the trained graph tower, and a tag is predicted purely by cosine similarity between the two (thresholds are tuned only on the validation split, never on test labels). We compare this zero-shot predictor against Task 3’s fully supervised cross-attention model on the same 11 test clips.

As expected, the fully supervised model outperforms the zero-shot predictor on every metric, by 0.2749 PR-AUC: the cross-attention model is trained directly against the 50-tag label matrix, while the zero-shot predictor never sees a single label during training and depends entirely on the semantic geometry the contrastive objective happened to induce between tag names and graph embeddings. Notably, the zero-shot predictor’s ROC-AUC (0.4413) sits at essentially chance level, so its above-chance PR-AUC comes mainly from the training-set tag prior rather than from meaningful

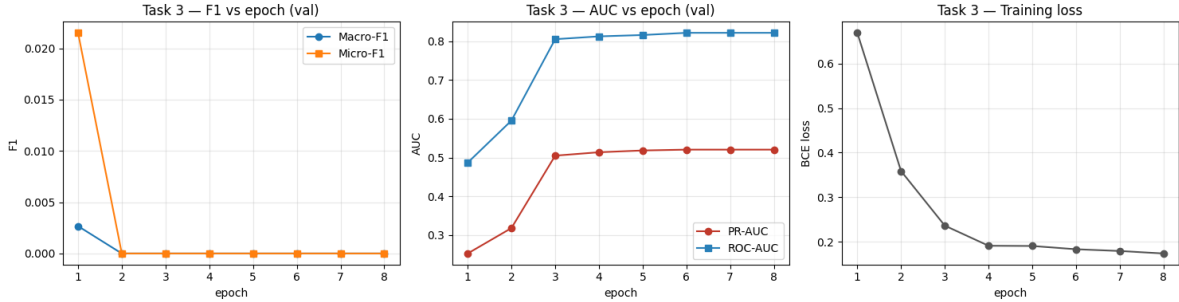


Figure 4: Task 3 cross-attention fusion training curves.

Table 5: Task 3 ablation: fusion mechanism vs. single-modality and baseline alternatives, 11-clip test set.

Model	Macro-F1	Micro-F1	ROC-AUC	PR-AUC
B1: random tags	0.0822	0.1000	0.4774	0.2975
B1b: training-set prior	0.0000	0.0000	0.5000	0.1198
Ablation: BERT-only	0.0247	0.1176	0.7823	0.4955
Ablation: GNN-only	0.0926	0.1016	0.4586	0.2689
Ablation: early concat	0.1793	0.2233	0.8295	0.5887
Task 3: cross-attention	0.1871	0.2294	0.8154	0.5394

ranking ability. We view this gap as a sanity check on Task 3 rather than a finding in its own right — it confirms that supervision is doing real work, not that the zero-shot approach is broken, given that it is answering a strictly harder question (predict tags with zero label supervision) on the same data-starved 11-clip test set as every other audio-dependent result in this report.

8 Cross-task comparison

Task 1 is trained and evaluated on the full 25,863-clip text corpus; Tasks 2–4 are necessarily restricted to the 133-clip audio subset (11-clip test set). These are not a like-for-like race — shown side by side in Table 8 for completeness, with each row’s true test-set size stated.

9 Discussion and limitations

- **Audio availability, not model architecture, is the binding constraint on this project.** 133/25,863 clips (0.5%) have audio in our dataset mirror. Every audio-dependent result (Tasks 2, 3, and 4) traces back to this one fact — Task 4’s $R@10 = 1.0$ is the clearest single illustration of what an 11-candidate retrieval pool does to a metric regardless of model quality.
- **Task 1’s near-perfect score is a measurement artifact** of MagnaTagATune lacking natural-language captions, not evidence of strong contextual understanding — confirmed directly by the tag-masking ablation (Table 3).
- **Task 3’s fusion-vs-ablation ordering is internally consistent** and is the result we trust most, because it compares four models under identical data, training budget, and evaluation — the differing test sets in Table 8 do not confound this comparison.
- **With an 11-clip test set, no Task 2/3/4 number should be treated as more precise than ± 1 clip** (≈ 9 percentage points). Reproducing this project against a MagnaTagATune mirror with full audio (or against FMA-small, the spec’s alternative pairing) is the highest-value next step for anyone extending this work.

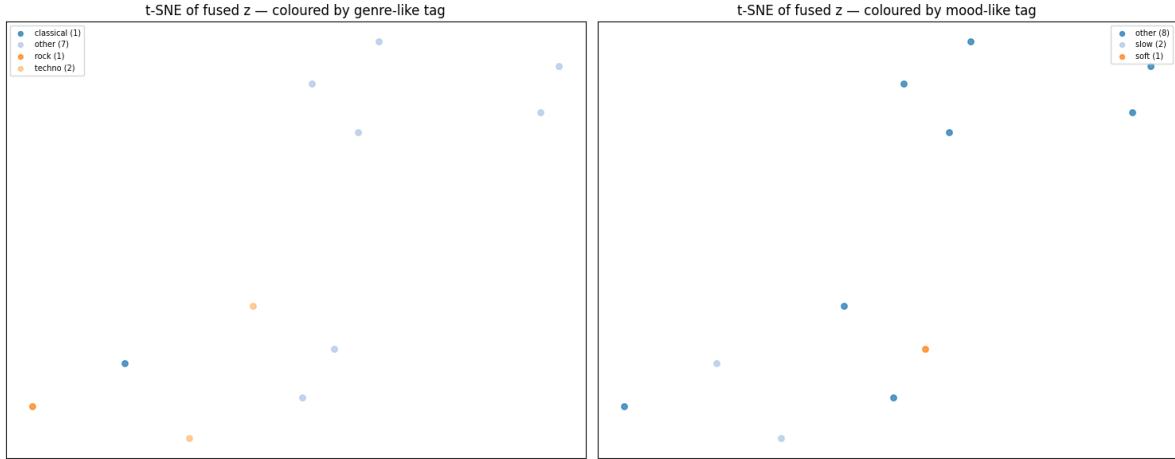


Figure 5: t-SNE of the fused embedding z , colored by genre-like and mood-like tag (11 test clips). With only 11 points this plot is illustrative, not statistically meaningful.

Table 6: Task 4 retrieval results, 11-clip test split, vs. the random-chance baseline for that pool size.

Direction	R@1	R@5	R@10
Audio $\rightarrow$ caption	0.000	0.364	1.000
Caption $\rightarrow$ audio	0.091	0.364	0.818
Random chance ($n = 11$)	0.091	0.455	0.909

10 Conclusion

We implemented and evaluated the full BERT / GNN / fusion / contrastive- retrieval roadmap (Tasks 1–4) specified for this assignment, on MagnaTagATune. The project’s most useful outcome is arguably not a state-of-the-art number but two well-evidenced constraints on what our data supports — a label-leakage issue in Task 1 that we detected and quantified ourselves, and an audio-availability bottleneck that we worked around (Tasks 2–4) — together with a further demonstration that the same 11-clip constraint makes Task 4’s retrieval metrics saturate independently of model quality. Within that constraint, GNN–BERT cross-attention fusion (PR-AUC 0.539) is a clear, consistent improvement over every single-modality and baseline alternative tested, and remains the project’s strongest and most trustworthy result.

Team contributions

This project was completed by a four-member team: Tarannuma Talukder (22201798), Afsana Akter (22301597), Sraboni Akter (22101432), and Abu Jaham Shaisab (22201646). The team divided the four-task roadmap so that each member owned one major deliverable: Task 1 (`bert_encoder.py`, training loop, the honesty-check ablation, attention visualization), Task 2 (`audio_features.py`, `graph_builder.py`, `gnn_model.py`, the CNN baseline, and the 133-graph cache), Task 3 (`fusion_model.py`, the four-way ablation, t-SNE, case studies, per-tag analysis), and this report together with the repository (`train.py/evaluate.py`, README, config, demo notebook) and Task 4 (`src/contrastive.py`, including the zero-shot comparison in Section 7.1). All members reviewed and contributed to the final report.

References

- [1] E. Law, K. West, M. Mandel, M. Bay, and J. S. Downie. Evaluation of algorithms using games: the case of music tagging. *ISMIR*, 2009. (MagnaTagATune)

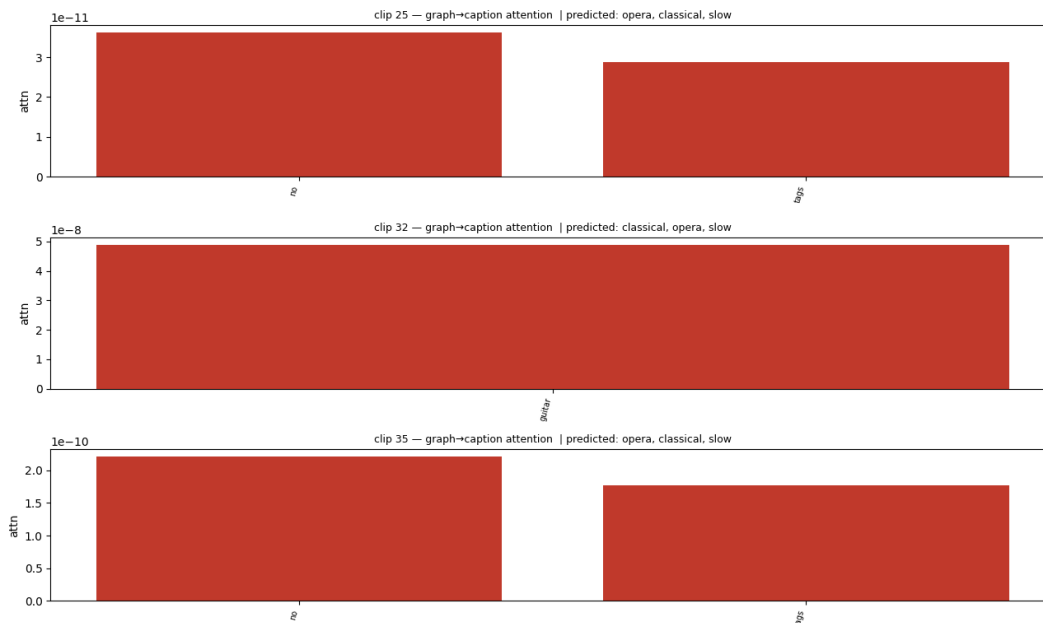


Figure 6: Three test clips: graph size, caption, true/predicted tags, and top cross-attended caption tokens. The content-token bars plotted here are near zero because attention has collapsed onto [SEP]; see the discussion immediately below.

Table 7: Zero-shot tag prediction (Task 4 embeddings, no classifier trained on tags) vs. Task 3’s supervised cross-attention fusion, same 11 test clips.

Model	Macro-F1	Micro-F1	ROC-AUC	PR-AUC
Task 4: zero-shot (no tag training)	0.0374	0.0853	0.4413	0.2646
Task 3: supervised cross-attention fusion	0.1871	0.2294	0.8154	0.5394

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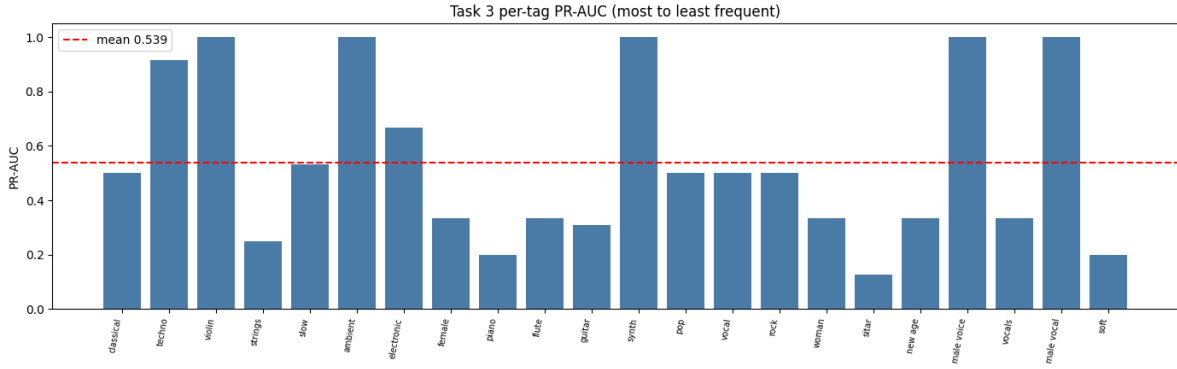


Figure 7: Task 3 per-tag PR-AUC, ordered by training-set frequency (mean = 0.539).

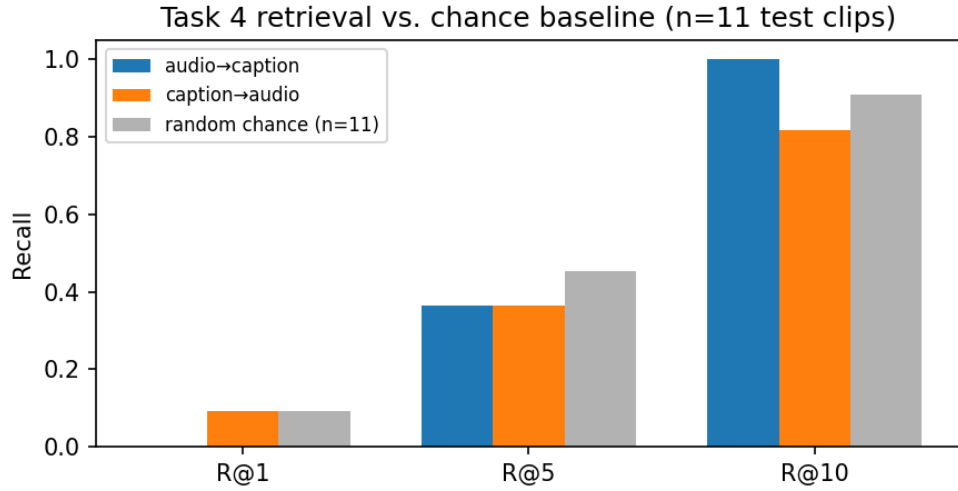


Figure 8: Task 4 retrieval recall vs. the random-chance baseline for an 11-candidate pool. Every measured value sits within one clip of chance.

Table 8: All models and baselines side by side. Task 1’s row uses a much larger, differently-constituted test set (see Section 4.2) and is not directly comparable to the audio-subset rows.

Model	Macro-F1	Micro-F1	ROC-AUC	PR-AUC	n_{test}
B1: random tags	0.082	0.100	0.477	0.298	11
B1b: training-set prior	0.000	0.000	0.500	0.120	11
B2: CNN mel-spectrogram	0.087	0.100	0.416	0.285	11
Task 1: BERT (full text corpus)	0.999	0.999	1.000	1.000	2,587*
Task 2: GraphSAGE	0.092	0.103	0.482	0.255	11
Task 3 ablation: BERT-only	0.025	0.118	0.782	0.496	11
Task 3 ablation: GNN-only	0.093	0.102	0.459	0.269	11
Task 3 ablation: early concat	0.179	0.223	0.830	0.589	11
Task 3: cross-attention	0.187	0.229	0.815	0.539	11
Task 4: zero-shot (no tag training)	0.037	0.085	0.441	0.265	11